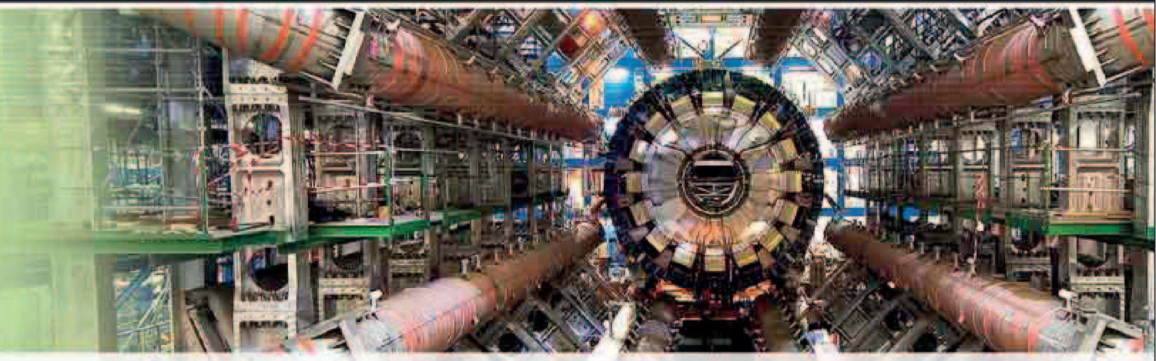
27 MILES THE APPROXIMATE CIRCUMFERENCE OF THE LHC



The ATLAS detector in the Large Hadron Collider is surrounded by eight huge electromagnets, which deflect the particles produced during collisions to that their mass and charge can be determined.

IN FEBRUARY 2008, A SECURE SEED BANK WAS OPENED ON the island of Spitsbergen in the Norwegian archipelago of Svalbard in the Arctic Ocean. Seed banks (see 2000) store seeds that could boost the populations of important plant species should climate change, wars, or natural disasters threaten their survival. However, seed banks themselves can also be vulnerable: in 2004, war claimed a seed bank at Abu Ghraib in Iraq, while floods destroyed one in the Philippines in 2006 after a typhoon hit. The Svalbard Global Seed Vault has the **capacity to hold up to 4.5 million seeds** and is embedded in an icy mountain in a remote location, providing more secure storage.

Icy seed storage
Situated on the Norwegian island of Spitsbergen, the Svalbard Global Seed Vault can hold up to 4.5 million seeds at freezing 0ºF (–18ºC).

The world's largest and most powerful particle accelerator, the **Large Hadron Collider (LHC), was turned on for the first time** in September 2008. The LHC is situated in a huge, circular, underground tunnel that straddles the border between France and Switzerland, at the European Organization for Nuclear Research (CERN). When the LHC is in operation, beams of protons (or for some experiments, ions) circle at extremely high speeds and are made to collide inside detectors. The **energy of the collision creates new particles**, and the detectors record the tracks of these particles as they hurtle out in all directions. These tracks are then analyzed by a network of powerful computers to look for evidence of specific types of particle—in particular, the **Higgs boson**. This particle is associated with the Higgs field, which, according to theory, is responsible for giving particles mass. Scientists have since revealed convincing evidence for the existence of the Higgs boson, and therefore the Higgs field (see 2011–12).

Several hundred extrasolar planets (planets outside our Solar System) had been found since the first example was detected (see 1992–93). In March 2009, NASA launched the **Kepler space observatory to find Earth-sized planets** around relatively nearby stars and to help astronomers estimate what proportion of stars have Earth-like planets. By 2012 it had discovered over 2,000 candidates.

In May 2009, Norwegian palaeontologist Jørn Hurum (b.1967) unveiled a remarkable specimen: an almost complete, fossilized skeleton of a **previously unknown species of lemur-like animal** that lived 47 million years ago. The specimen was claimed to be a transitional fossil—a **missing link** between lower primates, such as lemurs, and higher primates such as monkeys, apes, and humans. The fossil was

Darwinius masillae
Nicknamed "Ida," this fossil of a female is the only known example of Darwinius masillae, *quite possibly a direct human ancestor that lived 47 million years ago.*

February 26, 2008 Opening of the Svalbard Global Seed Vault on the island of Spitsbergen, Norway

September 10, 2008 The Large Hadron Collider begins operation

September 20, 2008 A consortium led by Google releases the **first version of the Android operating system**

October 7, 2008 Start of the Human Microbiome Project to investigate microbes that live on humans

March 7, 2009 Launch of **Kepler space observatory** to search for Earth-like extrasolar planets

May 19, 2009 Scientists announce the **discovery of *Darwinius masillae*,** a human ancestor that lived 47 million years ago

July 2009 Scientists at ETH Zurich announce the creation of a **transistor made from a single molecule**

August 2009 Amino acids found in material **from a comet**